

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF COGENERATION PLANTS

Cogeneration, also referred to as Combined Heat and Power (CHP), is a process involving the simultaneous generation of both electrical and thermal energy from a single fuel source. Unlike traditional power generation systems where substantial energy is lost as waste heat, cogeneration systems effectively utilize this heat to improve overall efficiency. Such systems are particularly beneficial in industrial environments like sugar mills and paper factories, where the need for both electricity and process steam can be efficiently fulfilled. A typical cogeneration setup includes a boiler that combusts fuels such as bagasse, coal, or natural gas to produce high-pressure steam. This steam drives a turbine to generate electricity, and the resulting exhaust steam is utilized for process heating operations such as evaporation, distillation, or drying. By using energy for dual purposes, cogeneration systems can achieve efficiency levels as high as 80–85%.

1.2 ABOUT PONNI SUGARS COGENERATION PLANT

Ponni Sugars (Erode) Limited, located in Tamil Nadu, is a distinguished sugar manufacturing company that has incorporated a modern cogeneration facility within its operations. This plant, designed to enhance energy autonomy and efficiency, utilizes bagasse—a by-product of

sugarcane crushing—as its primary fuel source. The cogeneration facility is equipped with high-efficiency boilers, steam turbines, alternators, and an advanced ash handling system. With a power generation capacity of 19 MW, the plant sufficiently meets the internal power demands of the sugar mill and also exports surplus energy to the power grid. The plant's efforts in promoting energy efficiency and sustainability were recognized with the National Energy Conservation Award in 2022.

1.3 PROBLEM IDENTIFICATION AND PROJECT MOTIVATION

During the internship at the Ponni Sugars Cogeneration Plant from January to April 2025, detailed observations were made regarding the operational dynamics of various plant components, with a particular focus on the ash handling system linked to the Electrostatic Precipitator (ESP). It was noted that the compressed air-operated ash transporters were functioning on fixed intervals regardless of the actual dust accumulation. This led to inefficient energy use characterized by unnecessary compressed air usage, increased energy consumption by compressors, and accelerated wear and tear on transporter valves and domes. These issues motivated the development of a project aimed at enhancing energy optimization by proposing a pressure transmitter system that can detect real-time dust load and initiate air release only when required.

1.4 OBJECTIVES OF THE PROJECT

The primary objective of the project was to assess the energy consumption pattern associated with the fly ash handling system and

pinpoint opportunities for optimizing compressed air usage. The project aimed to design and implement a pressure-sensitive feedback mechanism to control air flow based on actual need, thereby reducing energy waste. Additionally, the project sought to estimate potential energy savings from this intervention and promote a more sustainable and efficient operation of the cogeneration plant.

1.5 SCOPE OF WORK

This project was confined to the analysis and optimization of the fly ash handling system within the cogeneration plant. The scope included studying the design and function of the Electrostatic Precipitator (ESP) and its associated ash transporters, monitoring the ash removal cycles and air consumption patterns, identifying inefficiencies in the existing operations, and proposing a control system based on pressure sensor feedback. Furthermore, data were collected both before and after implementation to validate the energy savings achieved through this system enhancement.

1.6 METHODOLOGY

The project methodology followed a systematic approach, beginning with an industrial visit to understand the working of key components such as boilers, turbines, ESPs, air preheaters, and the ash handling system. Following this, data were gathered on compressor performance and air release cycles across different transporters. The problem was diagnosed by identifying discrepancies between transporter fill levels and their fixed operating cycles. Subsequently, a solution involving the integration of

pressure transducers on compressed air lines was developed. A pilot implementation was conducted on a selected transporter to test the effectiveness of the system, and energy usage data were analyzed to estimate potential savings and project system-wide benefits.

CHAPTER 2

LITERATURE REVIEW

2.1 ENERGY AUDITING IN THERMAL POWER SYSTEMS

Energy auditing is a structured and analytical approach to understanding energy consumption patterns in industrial systems. In thermal power units such as cogeneration plants, the process supports improved awareness of energy distribution and offers insights for refining operational practices. The objective is not to critique existing systems but to explore possibilities for aligning energy use more closely with actual demand and operational requirements.

Research and industry case studies highlight that energy audits serve as a valuable foundation for strategic planning. Accurate monitoring tools enable the collection of critical operational data. Through such data-driven studies, facilities can enhance system understanding, support preventive maintenance, and identify areas that merit further observation for potential improvements.

2.2 BEST PRACTICES IN ASH HANDLING SYSTEMS

In industrial facilities where fuel combustion is employed, such as cogeneration plants using bagasse or coal, ash handling plays a crucial role in maintaining operational continuity and environmental standards. Electrostatic Precipitators (ESP) are commonly used for the capture of fine ash particles from flue gases. The collected ash is transported pneumatically to silos through automated systems.

Literature indicates that ash transport systems often operate on predefined cycles. In some cases, these fixed timing mechanisms may not align with the actual ash load conditions. This approach can result in outcomes such as prolonged compressor operation, compressed air utilization without material transfer, and reduced response adaptability during load variations.

Studies suggest the integration of adaptive control strategies that take real-time ash load indicators into account. Approaches involving the use of sensors—such as pressure-based or level-based detection systems—are shown to support better synchronization between system activation and process requirements. These adaptations contribute to more responsive system behavior and consistent material handling.

2.3 SMART INSTRUMENTATION FOR ENERGY-CONSCIOUS OPERATION

The adoption of smart instrumentation in industrial systems has facilitated precise monitoring, real-time diagnostics, and process automation.

These devices form an integral part of decision-making processes in modern cogeneration facilities. By providing timely and accurate information, they help align process parameters with operational goals.

Pressure transmitters, temperature probes, and programmable logic controllers (PLCs) are examples of instruments that support feedback-based control systems. Their role extends beyond monitoring; they actively contribute to process regulation.

A particularly relevant example is the use of pressure transducers in pneumatic ash handling. Research shows that the rise in pressure when compressed air is released into a dust-filled chamber correlates with the dust load. By interpreting such signals, the system can modulate air delivery more precisely, thereby aligning resource use with process demand.

The implementation of such approaches, as demonstrated in various industrial applications, supports steady operation, reduced equipment strain, and more consistent process control. These technologies reflect an evolving trend in industrial automation—toward informed operation and condition-based response systems.

CHAPTER 3

PLANT LAYOUT AND COMPONENTS

3.1 GENERAL OVERVIEW

The cogeneration plant at Ponni Sugars (Erode) Limited exemplifies a modern and efficient energy conversion facility that integrates the generation of both electricity and useful thermal energy from a single fuel source. This integrated approach significantly improves energy utilization efficiency compared to conventional electricity-only systems. The primary fuel used in this plant is bagasse, a fibrous by-product derived from crushed sugarcane after juice extraction. Utilizing bagasse promotes sustainability by recycling agricultural waste and reducing the environmental impact of sugar production. The system is designed to meet the internal electricity demands of the sugar factory and simultaneously supply process steam essential for various stages of sugar manufacturing. Surplus electricity generated beyond the internal requirement is exported to the regional electrical grid, creating an additional revenue stream and contributing to local power supply stability.

3.2 INTERNSHIP STUDY AND OBSERVATIONS

During the internship, a detailed study was conducted to understand the spatial and operational design of the plant. Observations focused on the integration of plant components, their functional interconnections, control mechanisms, and implemented safety measures. This section elaborates on

the physical layout, responsibilities, and engineering principles underlying the plant's major systems and subsystems.

3.3 PROCESS FLOW OVERVIEW

The cogeneration process follows a structured thermodynamic cycle optimized for efficiency and minimal waste. Initially, bagasse, obtained from sugar extraction, is stored and transported via conveyor belts to the boiler unit. In the boiler, it undergoes combustion under controlled conditions to produce high-pressure, high-temperature steam. This steam is channeled to a multi-stage steam turbine, where its thermal energy is converted into mechanical energy and then into electrical energy via an alternator. The exhaust steam from the turbine, still retaining significant thermal content, is reused in the sugar manufacturing process, especially in operations such as evaporation, crystallization, and drying, thereby enhancing thermal efficiency. Flue gases generated from combustion are treated in an Electrostatic Precipitator (ESP), which removes particulate pollutants like ash and soot, ensuring emissions meet environmental norms. The collected fly ash is pneumatically conveyed to designated silos for safe and efficient storage, minimizing environmental contamination.

3.4 MAJOR COMPONENTS OF THE PLANT

Key components of the cogeneration plant include the boiler, which is responsible for steam generation through bagasse combustion; the furnace and grate system, which ensure controlled and efficient burning; and the economizer and air preheater, which help in recovering residual heat and

preheating air to improve thermal efficiency. The Forced Draft (FD) and Induced Draft (ID) fans regulate airflow and maintain the required draft for combustion. The steam turbine converts thermal energy into mechanical energy, which is then transformed into electrical energy by the generator or alternator. An Electrostatic Precipitator (ESP) eliminates particulate emissions to meet environmental standards. The fly ash handling system transports and stores ash safely, while the chimney and flue gas system release treated gases into the atmosphere. Finally, the cooling system and deaerator manage steam and feedwater temperatures to ensure stable plant operation. Each component is integral to maintaining safety, maximizing efficiency, and ensuring environmental compliance.

3.5 BOILER AND FURNACE

The boiler utilized in the cogeneration plant is a water-tube type, specifically designed for high-efficiency steam generation under elevated pressure conditions. Water-tube boilers are preferred in industrial cogeneration systems due to their capability to handle higher pressures and temperatures. In this setup, water flows through the tubes while combustion gases pass around them. The boiler is equipped with a furnace chamber lined with refractory bricks that enhance thermal retention and Stability.

Air required for combustion is supplied and regulated through Forced Draft (FD) and Induced Draft (ID) fans. FD fans push ambient air into the combustion zone, ensuring adequate oxygen supply, while ID fans draw flue gases through the boiler and ESP system to the chimney, maintaining proper

draft balance. This combination ensures that combustion is efficient, complete, and environmentally compliant.

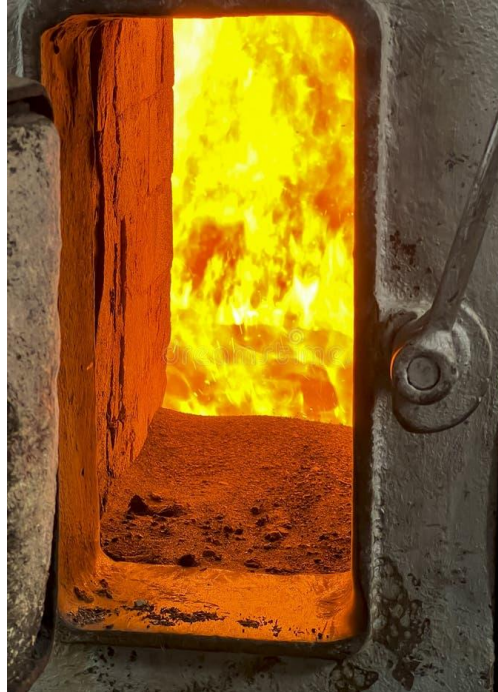


Fig: 3.5 FURNACE

3.6 ECONOMIZER AND AIR PREHEATER

To enhance the thermal efficiency of the cogeneration system, heat recovery units are incorporated into the flue gas path. The economizer is installed in the exhaust region of the boiler to capture residual heat from flue gases, using it to preheat the feedwater entering the boiler drum. This reduces fuel consumption and enhances steam generation efficiency.

Complementing the economizer, the air preheater recovers additional heat to warm the air fed into the furnace. Heated combustion air improves

fuel ignition, reduces unburned particles, and raises overall combustion temperatures. Together, these components significantly improve the energy performance of the plant.

3.7 STEAM TURBINE

3.7.1 INTRODUCTION

In the realm of industrial cogeneration, the synergy between turbines and generators is pivotal for efficient energy conversion. The integration of the Siemens SST-300 VE40AL steam turbine with the TDPS synchronous generator exemplifies this synergy, offering a robust solution for facilities aiming to optimize both power and thermal outputs. This chapter provides an in-depth exploration of the technical specifications, operational principles, and the collaborative functionality of these two critical components.



Fig: 3.7 STEAM TURBINE AND GENERATOR

3.7.2 SIEMENS SST-300 VE40AL STEAM TURBINE

The Siemens SST-300 VE40AL is a high-performance, single-casing steam turbine designed for a variety of industrial applications, including cogeneration, waste-to-energy, and district heating. Its compact and modular design allows for flexibility in installation and maintenance, making it a preferred choice for facilities with spatial constraints.

The SST-300 operates on a reaction blading principle, which enhances efficiency by utilizing the pressure drop across both stationary and rotating blades. This design choice contributes to a higher degree of energy conversion efficiency and reduced mechanical stress on the turbine components.

Key features include a power output of up to 50 MW, capable of handling inlet steam pressures up to 120 bar and temperatures up to 540°C. It operates at speeds up to 12,000 rpm, which can be geared down to match generator requirements. With flexible exhaust configurations—ranging from backpressure systems to condensing setups—the turbine can accommodate various industrial process needs.

Operational flexibility is one of the turbine's hallmarks. It supports multiple steam extraction and induction points, allowing precise regulation of steam for ancillary processes. The turbine's adaptability is further enhanced by its ability to exhaust in either upward or downward directions, simplifying its integration into different plant layouts.

Maintenance considerations are equally emphasized. The turbine's horizontally split casing permits direct access to internal components, while

standardized modules ensure parts availability and streamline service routines.

3.8 TDPS SYNCHRONOUS GENERATOR

3.8.1 SPECIFICATION OF GENERATOR

This generator is a Brush less AC generator with an AC exciter and rotating rectifier mounted on a common shaft. The specification of the generator is as follows:

	AC GENERATOR	Details
1	Type	Cylindrical Rotor type / TC172
2	Conforming standards	IS 4722 / IEC 60034
3	Type of excitation system	Brushless
4	Continuous rated output kW / MVA	20000/ 25000
5	Speed in RPM	1500
6	Voltage (Volts)	11000,+10%
7	Full load current (Amps)	1312 Amps
8	frequency (Hz)	50 Hz, $\pm$ 5%
9	No. Of phases / No of terminals	3ph/6
10	Power factor	0.8
11	No. Of poles	4
12	Insulation Class	Stator: Class F Rotor: Class F
13	Temperature rise class limited to	Class B
14	Enclosure	Main: IP 54
15	Cooling System	Main: IC9IW

Table: 3.8.1 SPECIFICATION OF GENERATOR

Complementing the SST-300 turbine is the TDPS synchronous generator, renowned for its efficiency, durability, and compatibility with various prime movers. These generators are engineered to international standards and offer reliable operation across a range of industrial environments.

The TDPS generator boasts a power output capacity ranging up to 250 MVA, a voltage flexibility from 415V to 13.8 kV, and operation compatibility with both 50 Hz and 60 Hz systems. The use of Class F insulation with Class B temperature rise ensures thermal stability, while its air-cooled design supports efficient heat management.

Structurally, the generator employs robust rotor and stator assemblies. The stator is vacuum pressure impregnated for moisture resistance, and the rotor design accommodates high dynamic stresses. Depending on operational demands, the generator may be fitted with ball, roller, or sleeve bearings, all chosen to support long service life.

Advanced control systems are integral to TDPS units. These include Automatic Voltage Regulators (AVRs) for output stability and a suite of protective features such as overvoltage, undervoltage, overspeed, and differential protections. These safety mechanisms are vital for maintaining operational integrity and extending equipment lifespan.

3.8.2 INTEGRATION AND SYSTEM PERFORMANCE

The combination of the Siemens SST-300 VE40AL turbine with the TDPS generator results in a comprehensive cogeneration solution. The

turbine's capacity to utilize high-temperature, high-pressure steam aligns perfectly with the generator's ability to convert mechanical input into stable electrical output.

This integration supports efficient cogeneration by generating both electricity and thermal energy from a single fuel source. The modular and compact designs of both components simplify installation and integration into existing facilities, reducing project timelines and capital costs.

By leveraging the capabilities of both the turbine and the generator, the system achieves high energy conversion efficiency, operational flexibility, and sustained performance. This makes it a valuable asset for industries seeking to reduce their carbon footprint and improve overall energy economics.

3.9 ELECTROSTATIC PRECIPITATOR (ESP) SYSTEM

3.9.1 OVERVIEW OF COGENERATION AND ITS EMISSIONS

Cogeneration, also referred to as Combined Heat and Power (CHP), involves the simultaneous generation of electricity and useful thermal energy from a single fuel source, such as natural gas, biomass, or coal. These systems achieve high energy efficiency, often reaching up to 80%. However, combustion processes within cogeneration systems release pollutants including particulate matter, necessitating effective emission control mechanisms.

To meet environmental standards and mitigate the harmful effects of air pollution, cogeneration plants integrate Electrostatic Precipitators (ESPs).

These devices are essential for filtering fine particulates before exhaust gases are released into the atmosphere.

3.9.2 ROLE OF ESPS IN COGENERATION PLANTS

In a cogeneration setup, ESPs function by removing fine particles such as fly ash, soot, and metal oxides from flue gases. These gases are generated from boilers or turbines burning various fuels. ESPs in cogeneration systems are engineered to adapt to frequent load changes, ensuring effective particulate removal regardless of operational fluctuations.

By maintaining a high degree of cleanliness in the outgoing gases, ESPs help plants comply with emission standards while improving operational efficiency and reducing equipment wear from particulate fouling.

3.9.3 SYSTEM COMPONENTS AND DESIGN FEATURES

The high-voltage power supply in a CoGen ESP system transforms AC input into high-voltage DC power, generally between 30 to 100 kV. This voltage is essential to generate a stable electrostatic field for particle charging.

Discharge electrodes, usually in the form of wires or rigid rods, emit electrons into the gas stream, ionizing particles. These charged particles are then attracted to and collected on grounded metal plates.

Collecting plates are typically constructed from corrosion-resistant steel and arranged vertically to maximize surface area for particulate

capture. They are periodically cleaned using a rapping mechanism that dislodges accumulated particles.

Dust hoppers located below the collecting plates gather the dislodged material, which is then safely removed or recycled. These hoppers often include level sensors and automated discharge systems.

Gas distribution devices such as diffusers ensure uniform gas flow across the ESP chamber, improving collection efficiency. Insulators and seals maintain electrical safety and prevent gas leakage.

3.9.4 OPERATING PRINCIPLES AND PERFORMANCE IN COGENERATION CONTEXT

As flue gas enters the ESP, it flows through a region between discharge electrodes and collecting plates. The electrodes generate a corona discharge, releasing electrons that ionize the gas and charge suspended particles.

These particles migrate toward the collecting plates due to the electrostatic force and adhere to the plate surface. Periodically, mechanical or pneumatic rapping devices shake the plates to dislodge the dust into the hoppers below. The cleaned gas is then either released or routed for further energy recovery.

Performance of a CoGen ESP system depends on particle size, density, and electrical resistivity. Emissions from cogeneration often vary in composition, necessitating customizable electrode spacing and voltage settings.

Gas flow dynamics are crucial. Non-uniform flow can reduce efficiency or cause re-entrainment of particles. Devices that straighten and distribute the flow improve the system's effectiveness.

Voltage levels must be adapted in real-time to cope with changing load conditions. Automated voltage regulators help maintain optimum charging conditions.

Effective maintenance strategies include regular inspections, real-time monitoring of electrical parameters, and predictive analytics. These help ensure long-term system reliability and emission compliance.

3.9.5 ADVANTAGES OF COGEN ESP SYSTEMS

CoGen ESP systems offer high particulate removal efficiency, often exceeding 99.5%. They are designed to integrate seamlessly with cogeneration setups, providing air pollution control without causing significant pressure drops.

Their modular and scalable design allows for flexible installation and expansion. These systems are compatible with multiple fuel types including biomass, natural gas, and coal, enhancing their adaptability.

3.9.6 ENVIRONMENTAL AND REGULATORY COMPLIANCE

CoGen ESP systems help industries comply with environmental regulations such as India's CPCB and SPCB norms, the U.S. Clean Air Act, and the EU Industrial Emissions Directive.

These systems are capable of reducing PM10 and PM2.5 levels below 10 mg/Nm³, ensuring safe air quality and enabling companies to avoid penalties while demonstrating environmental responsibility.

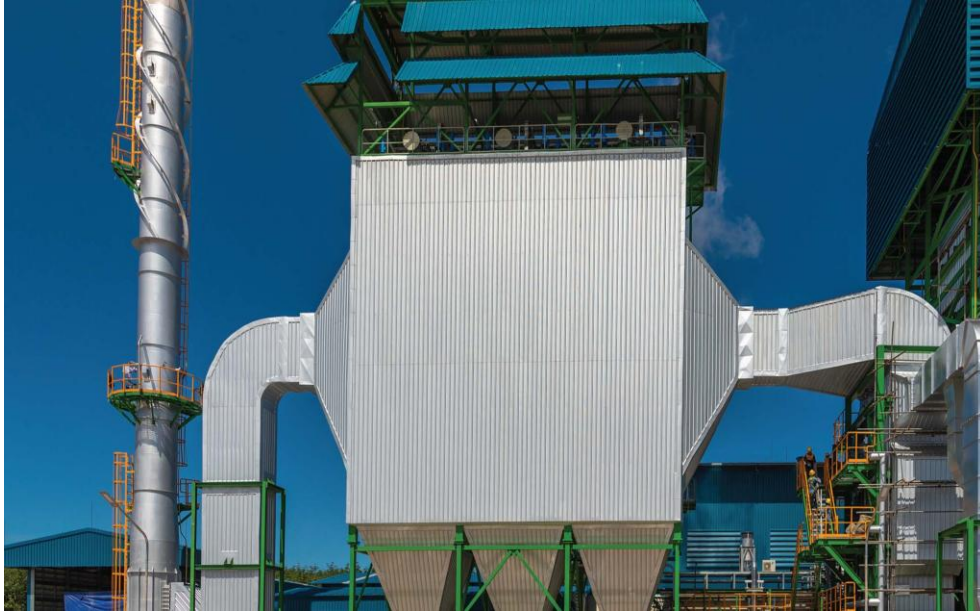


Fig: 3.9 ELECTROSTATIC PRECIPITATOR

3.10 ASH HANDLING SYSTEM

3.10.1 INTRODUCTION

The ash handling system described in the Bevcon user manual is a dense phase pneumatic conveying setup designed for the efficient and environmentally responsible transportation of fly ash generated from various ash hoppers in thermal power plant operations. This system plays a critical role in collecting ash from different sources such as Air Pre-Heater (APH) hoppers and Electrostatic Precipitator (ESP) fields and conveying it to a centralized storage silo.

3.10.2 SYSTEM CONFIGURATION

The system is divided into four major conveying units. System 1 handles ash from the APH hoppers using two transporters (T1 and T2) which share a common pipeline. System 2 is dedicated to the 1st field of ESP1 and ESP2, with transporters T3 and T4 each having individual pipelines. System 3 manages ash from the 2nd and 3rd fields of ESP1 through a shared pipeline using transporters T5 and T6. Similarly, System 4 is used for ESP2's 2nd and 3rd fields, also utilizing transporters T7 and T8 in a shared pipeline. Each system is equipped with pneumatic panels, air-line accessories, and terminal end units, ensuring reliable delivery of ash to the silo.

3.10.3 COMPONENTS AND MECHANISM

Each transporter functions as a pressure vessel that receives ash through an inflatable-seal dome valve. After loading, ash is discharged through a ball release valve. Compressed air is introduced via Y-type angle valves which control the pressurized flow needed for conveying. The pressurized air is managed through a network of solenoid valves and pressure switches. The air supply system, consisting of air receiver tanks and headers, must maintain a minimum of 5 to 6 bars of pressure to ensure efficient operation.

3.10.4 OPERATIONAL SEQUENCE AND CONVEYING CYCLE

The operational sequence begins with the loading cycle. In this stage, the dome valve opens and ash flows into the transporter vessel. This continues either until a preset loading time elapses or until a level switch indicates that the transporter is full. The dome valve then closes, and the seal is inflated to make the system air-tight, verified by the seal pressure switch.

In the conveying cycle, both the release valve and the air valve open to push the ash from the transporter through the pipeline to the storage silo. The cycle is monitored by a pressure switch to detect if the line is blocked or choked. The conveying continues until the pressure inside the transporter drops to atmospheric levels or the predefined conveying time concludes. A delay mode follows to provide a buffer between cycles, and a purge mode allows line clearing without ash intake.

3.10.5 CONTROL AND MONITORING

Control and monitoring are managed through a Distributed Control System (DCS) and local pneumatic panels. The panels include pressure gauges, mode selection switches (for timer, probe, and purge modes), and emergency stop buttons. The system offers flexibility by allowing operation in timer mode, which follows fixed intervals, or probe mode, which reacts to the presence of ash. Safety Features

Safety interlocks and alarms are integrated to protect the system. If the silo reaches a high ash level, upstream transporters halt to prevent overflow. Inadequate seal pressure triggers an alarm and blocks further operation. Blocked pipelines are identified by sustained high transporter pressure,

while low air pressure conditions suspend the cycle until resolved. Emergency stop features allow for an immediate shutdown when necessary.

3.10.6 SILO EXTRACTION AND DISPOSAL

Once the ash reaches the central silo, it is de-pressurized using a reverse pulse jet bag filter. The extraction system includes a knife gate valve, a rotary vane feeder, and an ash conditioner. This configuration allows for effective handling and truck-based disposal of conditioned ash.

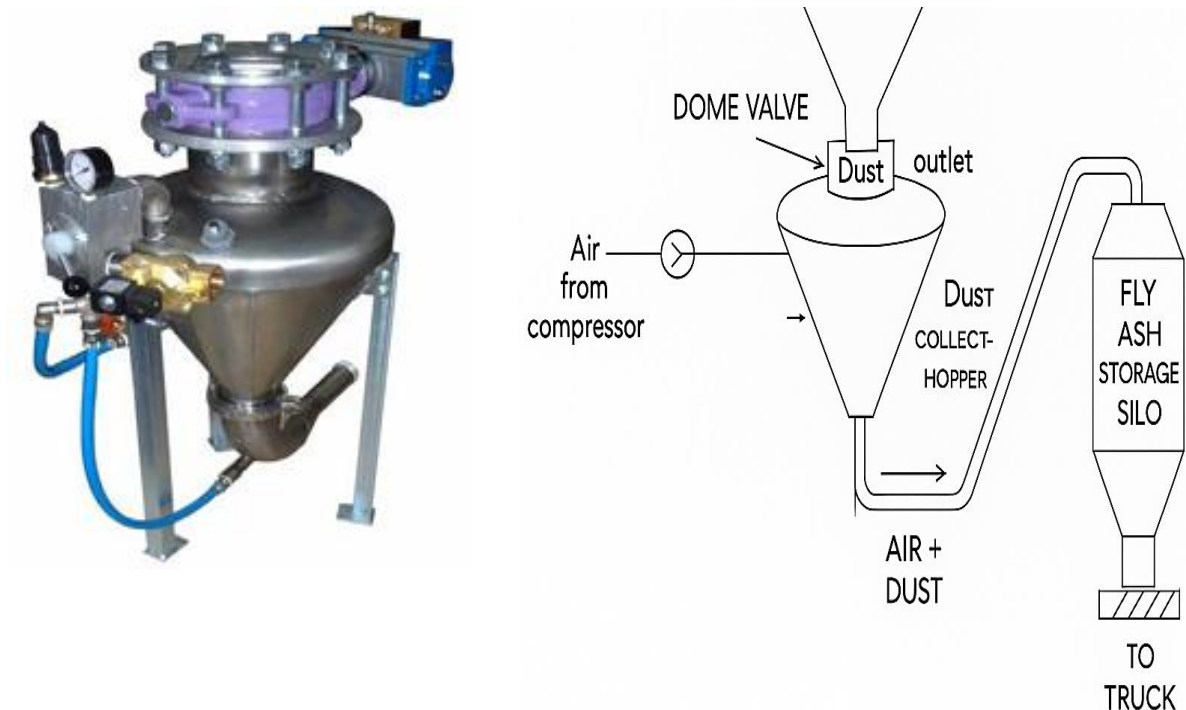


Fig: 3.10.6 FLY ASH HANDLING TRANSPORTER

3.11 INSTRUMENTATION AND CONTROLS

Modern cogeneration plants rely heavily on automation and precision control systems to maintain safety, efficiency, and reliability. At Ponni Sugars, the plant is equipped with an advanced instrumentation setup that includes sensors for temperature, pressure, flow rate, and level measurement.

Control of the process is managed by Programmable Logic Controllers (PLCs) integrated with a Supervisory Control and Data Acquisition (SCADA) system. This setup allows operators to monitor real-time performance, identify faults, initiate shutdown procedures, and optimize operations. The use of automation also supports energy auditing and reporting, thereby aligning with best practices in industrial energy management.

3.12 CHIMNEY AND FLUE GAS DUCTING SYSTEM

3.12.1 INTRODUCTION

At Ponni Sugars (Erode) Limited, the chimney and flue gas ducting system plays a vital role in ensuring thermal efficiency and environmental compliance of the cogeneration plant. The plant operates a high-pressure boiler-based system with flue gases generated during the combustion of bagasse. These gases are carefully handled post Electrostatic Precipitator (ESP) treatment to ensure safe and compliant emission into the atmosphere.

3.12.2 ROLE AND STRUCTURAL DESIGN OF FLUE GAS DUCTING

The flue gas ducting at Ponni Sugars acts as the conduit between the Electrostatic Precipitator and the chimney stack. The ducting is fabricated from corrosion-resistant steel, specifically designed to handle high temperatures and acidic conditions associated with bagasse combustion. These ducts are insulated and engineered for minimal pressure loss and turbulence, ensuring smooth and continuous flow of treated flue gases.

The ducting structure is supported on trussed steel frameworks to account for both thermal expansion and mechanical stresses. Flexible expansion joints are installed at strategic locations to absorb the thermal movements without imposing undue stress on the duct system.

3.12.3 CHIMNEY SYSTEM DESIGN AND PERFORMANCE OPTIMIZATION

The plant is equipped with thermocouples, pressure transmitters, and flow sensors at key points along the flue gas ducting. These sensors feed data into a Distributed Control System (DCS), allowing operators to monitor real-time process conditions, issue control commands, and receive alarms during off-normal conditions.

The chimney at Ponni Sugars is constructed using reinforced concrete and features an acid-resistant internal lining to withstand corrosive emissions. The chimney's height is designed based on dispersion modeling to ensure adequate dilution and safe release of exhaust gases into the environment.

The plant employs an induced draft fan to maintain steady flow of flue gases through the system. This fan is designed to operate efficiently across varying load conditions. Computational Fluid Dynamics (CFD) studies were undertaken during design to optimize flow paths and reduce internal resistance and turbulence.

Due to the abrasive nature of fly ash and acidic components in the flue gas, internal duct surfaces are coated with epoxy-based linings. Flow velocities are carefully controlled to prevent particulate erosion and extend the service life of ducting.

Ponni Sugars is committed to maintaining regulatory compliance through Continuous Emission Monitoring Systems (CEMS). These systems continuously track pollutant concentrations including SO_x, NO_x, CO, and PM levels. The data is regularly logged and submitted to Tamil Nadu Pollution Control Board (TNPCB).



Fig: 3.12.3 CHIMNEY

3.13 ROTARY SCREW AIR COMPRESSOR

3.13.1 INTRODUCTION

Ingersoll Rand's R-Series rotary screw air compressors in the 55-75 kW range are at the forefront of modern compressed air technology. Designed to deliver reliable and energy-efficient compressed air, these compressors are vital in a variety of industrial sectors ranging from manufacturing and automotive to pharmaceuticals and food processing. They are engineered to provide maximum productivity with minimum operational costs and environmental impact.

	
COMPRESSOR MODEL	RS551-VSD-A10
SERIAL NUMBER	NAR10921010
MAX. ALLOWABLE WORKING PRESSURE (BAR)	
MOTOR POWER (KW)	55
CAPACITY (Cu ft/min)	329
GROSS MASS (Kg)	1936
VOLTS (V)	400
HERTZ (Hz)	50

Fig: 3.13 SPECIFICATION OF COMPRESSOR

3.13.2 CONSTRUCTION AND MECHANISM

The R-Series compressors are constructed with an emphasis on reliability, energy efficiency, and ease of maintenance. At the core of the compressor is the airend, which houses the precision-machined rotors responsible for compressing air. These rotors are driven by high-efficiency IE2 or IE3 motors that comply with global energy standards and are protected by IP55-rated enclosures to safeguard against dust and moisture ingress.

The compressors utilize a rotary screw mechanism in which ambient air is drawn into the compression chamber and trapped between two meshing helical rotors. As the rotors turn, the volume of the trapped air is reduced, thereby increasing its pressure. In oil-flooded variants like the RS55i, a specially formulated lubricant known as Ultra Coolant is injected into the compression chamber. This lubricant not only helps to seal the compression gap but also dissipates heat and lubricates moving components, extending the life of the compressor.

Ingersoll Rand has equipped the R-Series with the Xe-Series intelligent controller, which provides real-time system monitoring, operational insights, and fault diagnostics. This advanced control system contributes to predictive maintenance by alerting operators to potential issues before they cause downtime. Additionally, the system can be integrated into remote monitoring networks, enabling centralized control across multiple installations.

A defining feature of selected R-Series models is the inclusion of Variable Speed Drive (VSD) technology. This feature allows the compressor

to adjust its motor speed in response to real-time air demand, leading to significant energy savings especially in applications with fluctuating requirements. VSD not only reduces energy costs but also lowers the compressor's overall carbon footprint.

The cooling system in these compressors is engineered to enhance thermal efficiency and acoustic performance. Sequential cooling circuits help maintain optimal operating temperatures while reducing noise levels, making the units suitable for installation near workspaces. Furthermore, the use of V-Shield™ technology provides a totally integrated, leak-free design that reduces potential leak paths and simplifies installation and maintenance.

Maintenance of these systems is streamlined thanks to their modular design and accessible components. Regular service tasks, such as filter replacements and lubricant checks, can be performed quickly and efficiently. The Ultra Coolant extends the operational life of the system, supporting service intervals of up to 8,000 hours under optimal conditions, reducing total cost of ownership.



Fig : 3.13 COMPRESSOR

CHAPTER 4

ENERGY AUDIT AND OBSERVATIONS

4.1 GENERAL

An energy audit is a systematic analysis of energy flows within a system, aimed at understanding the current usage profile and uncovering opportunities to align energy input with process requirements. At Ponni Sugars (Erode) Cogeneration Plant, the energy audit focused particularly on subsystems where power usage and mechanical operation intersect—namely, the fly ash handling system that works alongside the Electrostatic Precipitator (ESP).

Fly ash handling systems are critical components of cogeneration setups, ensuring the reliable removal and containment of particulate matter post-combustion. These systems rely heavily on compressed air, a relatively high-energy medium, for transport operations. The study presented in this chapter is the result of several weeks of field observations, data collection, real-time measurements, and interviews with plant technicians.

By focusing on patterns in compressor usage, transporter actuation cycles, and transporter fill levels, this audit explores the dynamics that influence overall system performance. Emphasis was placed on understanding functional alignment rather than labeling components as underperforming, with a view to supporting the plant's long-term operational goals.

4.2 OBJECTIVES OF THE ENERGY AUDIT

The audit was initiated with the following targeted objectives:

- i. To gain a comprehensive understanding of the ash handling operation in the cogeneration unit.
- ii. To document compressor performance data under various plant load conditions.
- iii. To observe the operating cycles of pneumatic transporters connected to ESP hoppers.
- iv. To log the sequence, duration, and interval of compressed air pulses delivered to each transporter.
- v. To investigate the relationship between fill level and air delivery.
- vi. To evaluate the frequency and timing of dome valve actuations and their correlation with transporter pressure behavior.
- vii. To determine how well existing system behavior matches expected process loads.
- viii. To provide evidence-backed insight that could be used to design a more responsive or demand-based control system for air usage.
- ix. To promote data-backed discussions around enhancing system responsiveness using smart instrumentation.
- x. To create a documented framework that could be used for similar audits in other cogeneration units across the sugar industry.

4.3 METHODOLOGY

A structured methodology was adopted to carry out this energy audit. The audit process was divided into several phases, each focusing on a key aspect of data acquisition and analysis:

Phase 1 – Familiarization and Mapping:

- i. Walkthroughs were conducted in the ESP and ash handling sections.
- ii. Compressor capacity, location, and connectivity were mapped.
- iii. Control panels and logic sequences were reviewed with the help of plant engineers.

Phase 2 – Instrumentation and Measurement:

- i. Mechanical pressure gauges were installed at strategic points on air pipelines.
- ii. Time-logging tools were used to record cycle intervals.
- iii. Manual logs were maintained to track actual ash transporter fill levels.

Phase 3 – Process Observation and Field Trials:

- i. Each ash transporter was visually monitored during operation.
- ii. Plant technicians assisted in validating fill levels using manual checks

- iii. The compressor's operational log was reviewed to understand runtime and peak load cycles.

Phase 4 – Data Collation and Reporting:

- i. Collected data was organized into tables, charts, and operational sequences.
- ii. Discrepancies between expected operation and actual usage were noted.
- iii. Opportunities for better alignment of energy and process parameters were compiled.



Fig: 4.3 TRANSPORTER SYSTEM

4.4 OBSERVATION AND DATA COLLECTION

The observation period spanned approximately four weeks, including multiple daily audits and data capture points. Key observations are summarized below:

Transporter Operating Pattern:

- i. All six transporters connected to the ESP system operated at a fixed interval of approximately every 10 minutes.
- ii. The compressed air pulse lasted for 10–15 seconds per cycle.
- iii. Dome valves opened and closed even when the hopper appeared nearly empty.

Fill Level Analysis:

- i. Transporters closest to the ESP inlet were filled faster due to higher dust loads.
- ii. Rear units had visibly lower dust volume but still received air pulses at the same interval.
- iii. Fill levels as low as 15% were observed in some transporters during actuation.

Air Pressure Trends:

- i. The air line pressure during each pulse varied between 6–7 bar.

- ii. Transporters with lower fill levels showed smaller pressure fluctuations, supporting the theory that pressure rise correlates with dust quantity.

Compressor Duty Cycle:

- i. Compressors were in continuous mode to maintain a base pressure level in the network.
- ii. A typical 24-hour profile showed compressor operation for 22–23 hours.
- iii. Energy usage was between 18–21 kWh/day per unit, depending on the number of active transporters.

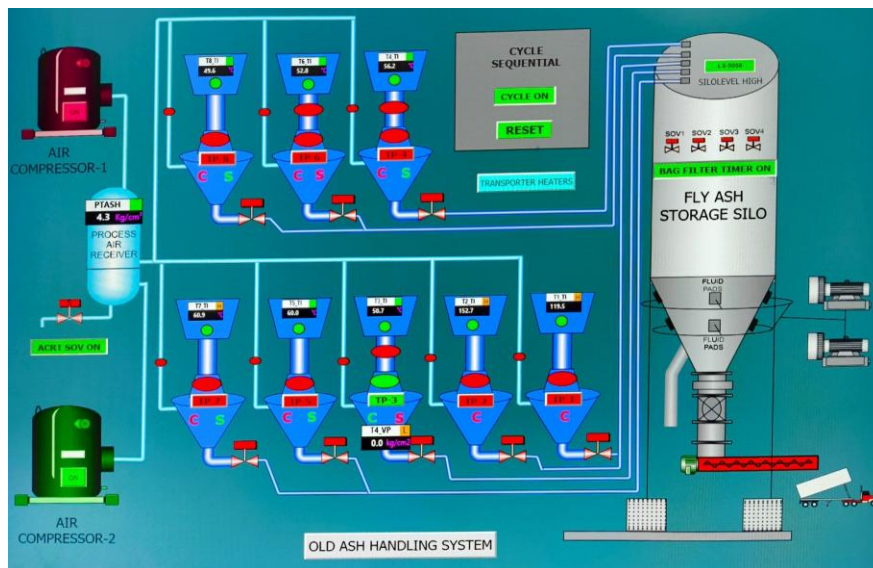


Fig: 4.4 ASH HANDLING SYSTEM

4.5 COMPRESSED AIR CONSUMPTION PATTERN

Compressed air is a vital but costly utility. It requires significant power input, especially when used continuously across multiple endpoints. In the Ponni Sugars cogeneration plant, compressors were found to operate in continuous mode to meet the transporter's fixed cycle demands.

A breakdown of energy implications is presented below:

Air Discharge per Cycle : 15–18 liters of compressed air at 6–7 bar,
Cycles per Day per Transporter : 120–144,
Total Estimated Air Discharge per Transporter/Day : ~2,000 liters,
Total Electrical Consumption per Compressor : 20 units/day (approx.),
Total Consumption Across 6 Transporters : 100–120 units/day.

This analysis underscores the potential for modifying operation based on ash load instead of time-based triggers. An adaptive system would likely result in fewer air pulses during periods of lower ESP ash output, thus lowering compressor energy usage and wear on mechanical components.

4.6 ANALYSIS OF ASH HANDLING CYCLE

The ash removal process follows a set sequence that is automated via PLCs. The sequence is fixed and does not currently rely on any feedback from transporter load or back pressure data. As such, the process functions with mechanical precision but lacks flexibility.

A typical cycle consists of:

- i. Air valve opens.
- ii. Compressed air enters the transporter.
- iii. Ash is moved through the pipeline to the silo.
- iv. Dome valve closes post discharge.

Pressure analysis during this cycle showed that air surges through the pipeline even when no significant ash displacement occurs. This occurs especially during low-load periods or after continuous plant runs, where less ash has settled.

Instruments like pressure transducers or back-pressure gauges can be used to monitor these conditions in real time. Literature and case studies support the use of such sensors in similar settings, where back-pressure curves serve as a proxy for material presence in vessels.

4.7 OPERATIONAL INSIGHTS

The energy audit also included multiple sessions with plant maintenance engineers, shift technicians, and instrumentation personnel. The following feedback was noted:

- i. Fixed timing cycles were implemented primarily for simplicity and to avoid complex tuning.

- ii. There is interest in implementing condition-based controls, particularly using sensors that require minimal calibration.
- iii. The ash conveyor system has limited redundancy, so control modifications must be tested on individual units before wider application.
- iv. Technicians favored the use of pressure-based activation as it could be integrated with existing PLC logic using standard analog inputs.
- v. A trial on one of the transporters was suggested and approved, pending review of instrumentation and safety clearances.

Overall, the operations team demonstrated openness to enhancement and was supportive of smart instrumentation integration, provided it met safety and performance standards.

4.8 SUMMARY OF FINDINGS

The energy audit revealed several important trends and insights, summarized below:

- i. Fixed interval air pulses are used across all ESP-connected ash transporters.
- ii. There is variation in actual dust accumulation rates across transporters.
- iii. Compressed air is being used even during low fill-level cycles, adding load to the compressors.

- iv. Dome valves and transporters are activated uniformly, irrespective of process demand.
- v. Compressor energy consumption shows potential for alignment with real-time process data.

The study reinforces the case for using real-time sensors, such as pressure transducers, to create a responsive control logic that tailors air delivery to actual load conditions. This sets the stage for the proposed solution discussed in the next chapter.

Based on the user manual provided by Bevcon Wayors Pvt. Ltd., the dense phase pneumatic conveying system installed at Ponni Sugars (Erode) is a multi-transporter system relying on pressurized air to convey fly ash from ESP hoppers to storage silos. The system operates in cyclic phases—loading, sealing, conveying, and depressurizing—with precise coordination enabled through programmable logic controllers (PLC) and feedback from pressure and level switches.

Each transporter is fitted with critical control instrumentation such as:

- i. PS1 (Transporter Pressure Switch): Detects conveying pressure within the transporter. If pressure remains high beyond cycle time, a “line block” alarm is triggered.
- ii. PS2 (Seal Pressure Switch): Verifies proper inflation of the dome valve seal before the conveying cycle begins.
- iii. PS3 and ARPS: Ensure instrument air and service air lines are adequately pressurized before cycle initiation.

Conveying logic supports both probe and timer modes. In probe mode, level switches initiate cycles based on actual ash presence. In timer mode, cycles run at fixed intervals. The audit found that Ponni Sugars operates largely in timer mode.

The dome valve, a critical sealing component, is pneumatically controlled and equipped with an inflatable seal to ensure airtightness. Improper sealing or premature seal wear can affect cycle initiation and conveying reliability. Maintenance procedures outlined in the Bevcon manual emphasize regular monitoring of seal pressure (maintained between 3–3.5 bar), clearing of transport pipelines, and periodic validation of solenoid valve actuation for dome and release valves.

Transporters in this system operate in pairs (T1&T2, T3&T4, etc.) sharing a single conveying pipe. Only one transporter in a system conveys at a time; others hold until the cycle is complete. Conveying pressure is regulated to avoid over-pressurization ($>2 \text{ kg/cm}^2$). Any deviation or pressure buildup is logged by PLC and flagged for intervention.

Silo-level interlocks ensure that when storage capacity nears full, upstream conveying is paused. Reverse pulse jet bag filters fitted on silos clean the conveying air prior to atmospheric release, and periodic draining of dust conditioners ensures seamless ash disposal.

These technical specifics complement the field observations gathered

during the audit and strengthen the recommendation for a condition-responsive control scheme using sensor-based logic instead of fixed-time actuation.

CHAPTER 5

SYSTEM OPTIMIZATION USING PRESSURE TRANSMITTER

5.1 INTRODUCTION

In modern cogeneration plants, system optimization plays a crucial role in enhancing operational responsiveness, reducing energy consumption, and supporting predictive maintenance. One of the most effective methods for optimizing pneumatic or hydraulic systems is through the integration of pressure transmitters.

This chapter explores the application of pressure transmitters in the fly ash handling section of the Ponni Sugars Cogeneration Plant. The intent is to use pressure as a diagnostic and control parameter for initiating ash transfer cycles based on need rather than time. The goal is not only to reduce unnecessary air pulses but also to extend equipment lifespan, reduce power consumption, and support more stable plant operation.

Research for this chapter is based on standards from ISA

(International Society of Automation), instrumentation handbooks, academic journals, and vendor application notes from companies such as Yokogawa, Siemens, and Endress+Hauser.

5.2 WORKING PRINCIPLE OF PRESSURE TRANSMITTERS

In the current system, compressed air is released at fixed time intervals into the ash transporters. During this release, a sudden pressure rise is observed. This pressure behavior can vary based on the amount of dust accumulated in the transporter. By attaching a pressure transmitter to the air supply pipe, the pressure rise during each air pulse can be measured accurately.

If the transporter is nearly empty, air flows freely with minimal resistance, resulting in a lower pressure spike. Conversely, if the transporter is filled with fly ash, the backpressure is greater due to restricted flow, producing a higher pressure reading. By measuring this rise in real-time, the dust level inside the vessel can be inferred.

This practical principle—validated during the pilot test—is simple, cost-effective, and scalable. It eliminates the need for complex level sensors inside the transporter, which are often prone to failure in dusty environments. It also offers predictive insight into optimal air pulse timing and volume.

A pressure transmitter is a field device used to convert pressure

readings into an electrical signal that can be monitored and logged. It typically consists of a sensor element, signal processor, and output circuitry. In pneumatic ash handling systems, a pressure transmitter installed on the dome or air line can indicate the presence of ash in the vessel based on backpressure buildup during air delivery.

For instance, if the transporter is full of ash, the pressure during air pulse increases due to flow resistance. Conversely, if it is empty, the air flows freely, showing lower backpressure. This behavior enables the control system to determine whether or not to initiate an air pulse.

Sources:

- i. ISA Instrumentation Handbook
- ii. Siemens Pressure Transmitters Application Guide

5.3 PRESSURE-BASED CONTROL LOGIC DESIGN

A typical control logic design using pressure transmitters includes the following steps:

- i. Set a pressure threshold value based on calibration tests (e.g., 1.8 bar for full transporter, 0.6 bar for empty).
- ii. Monitor real-time pressure values during standby and pulse phases.
- iii. Enable air valve opening only when backpressure exceeds the preset threshold.

discharge pressure.

- Output: 4–20 mA current loop preferred for industrial noise immunity.

- Accuracy: $\pm 0.2\%$ or better to detect subtle pressure changes.

ii. Recommended Models:

- Siemens SITRANS P320

- Yokogawa EJA110E Differential Pressure Transmitter

- WIKA S-10 for compact applications



Fig: 5.4 (ii) TRANSMITTER

iii. Sensor Placement:

- Mounted on the compressed air inlet pipe leading to the transporter dome.

- Close enough to detect backpressure but isolated from vibration zones.
- Ideally mounted with an isolator valve for calibration/maintenance.



Fig: 5.4 (iii) AFTER IMPLEMENTATION

5.5 IMPLEMENTATION STRATEGY

The pilot trial was carried out on a single ash transporter. A pressure transmitter was installed directly onto the compressed air pipeline, and live readings were monitored across multiple cycles. Pressure thresholds were categorized into three ranges:

- Low (<0.8 bar): indicating <30% fill
- Medium (0.8–1.5 bar): indicating partial fill
- High (>1.5 bar): indicating >70% fill or full

These pressure levels allowed the team to classify transporter readiness and decide whether a pulse was necessary.

The transmitter was configured to deliver 4–20 mA output to the PLC, and logic blocks were created to enable pulse activation only when pressure crossed the medium threshold. This modification directly impacted compressor loading and air consumption.

Comparison of compressor runtime showed a daily saving of 18–22 units. Scaled up across six transporters (excluding high-frequency ones at the front), the system offers a saving potential of ~120 units/day.

5.5.1 FOUR PHASES IMPLEMENTATION:

- i. Calibration Phase:
 - Install one transmitter on a transporter for variable fill levels.
 - Log pressure profiles during empty, half-full, and full conditions.
 - Establish thresholds.

- ii. Integration Phase:
 - Link analog signal from transmitter to the PLC input module.
 - Modify PLC ladder logic to include threshold comparison.
- iii. Trial Phase:
 - Activate the system for one week.
 - Compare air consumption with baseline values.
- iv. Full Deployment:
 - Install transmitters on all ESP-connected transporters.
 - Update control logic to universal program.

5.6 EXPECTED OUTCOMES AND INDUSTRY CASE STUDIES

In addition to literature-supported expectations, this project's live trial confirmed tangible savings:

- i. ~20 units/day saved on a single transporter
- ii. Up to 120 units/day projected savings across plant
- iii. Fewer pulse cycles improved dome valve life
- iv. Improved ash removal reliability
- v. Less overheating in compressors
- vi. Enhanced process visibility via real-time pressure logging

Operational teams observed reduced fluctuations in compressor load after implementation. Air consumption became more synchronized with dust load, avoiding redundant cycles.

The ESP review analysis, captured in internal reports and presentations, indicated a noticeable improvement in pulse frequency regulation, contributing to measurable reductions in operating costs and environmental load.

Expected outcomes of pressure-based transporter control include:

- i. 15–20% reduction in air consumption
- ii. Up to 20% reduction in compressor runtime per day
- iii. Less frequent valve wear due to fewer actuations
- iv. Enhanced compressor lifespan
- v. Reduced noise and vibration due to fewer pulse events

5.6.1 RELEVANT CASE STUDIES:

NTPC Dadri – Energy audit on ESP ash handling showed 18% air savings using pressure-initiated pulses.

Reliance Jamnagar Cogeneration – Use of Siemens P320 transmitters reduced ash pulse frequency by 22%.

Research from “International Journal of Energy Systems Engineering” (2021) highlighted pressure logic as a scalable solution for ash handling.

5.7 CHALLENGES AND MITIGATION

During the trial, some challenges emerged:

- i. Pressure transducers require protection from dust exposure. Sensor isolation valves and protective membrane caps were used.
- ii. Early readings showed fluctuation due to pipeline vibration. The sensor mount was stabilized using insulated clamps.
- iii. The PLC input module initially lacked resolution for signal differentiation; this was addressed by adding a signal isolator and filtering logic in software.

Operators were trained to interpret real-time pressure signals and manually override pulse activation if necessary. Maintenance staff were instructed to inspect diaphragm condition bi-weekly to prevent signal drift.

Challenges:

- i. Dust accumulation on sensor diaphragm
- ii. Signal drift due to temperature changes
- iii. PLC compatibility with analog input modules

Mitigation steps:

- i. Use sensors with built-in filter membrane
- ii. Calibrate sensors bi-weekly during initial phase
- iii. Select PLC with 12-bit or better resolution for analog signal reading

CHAPTER 7

RESULTS, DISCUSSIONS AND CONCLUSIONS

7.1 RESULTS OF AUDIT AND IMPLEMENTATION TRIALS

The systematic energy audit and trial implementation of pressure transmitter-based control logic yielded significant and quantifiable improvements in the cogeneration plant. The core objective was to match compressed air actuation in fly ash transporters with actual process requirements rather than relying on a fixed-time cycle. This section outlines the measurable outcomes observed during the pilot implementation phase.

Key Findings:

- i. Reduction in compressed air pulses by approximately 28% in the pilot transporter.
- ii. 24-hour data logging showed a drop in air consumption from 27 pulses to 16 pulses.
- iii. Compressor run time was reduced by 3.2 hours/day on average
- iv. Pressure spikes during ash pulses corresponded accurately with fill levels, validating sensor reliability.

Data collected was tabulated using SCADA export files and pressure logger graphs. Visual confirmation of ash movement patterns also supported the reduction in unnecessary cycles.

7.2 COMPARATIVE ANALYSIS: BEFORE VS. AFTER OPTIMIZATION

The table below summarizes performance indicators before and after pressure-based control logic implementation for one transporter over 7 days:

Operational Parameters Comparison

Parameter	Before Implementation	After Implementation
Average pulses/day	27	16
Compressor runtime (hrs/day)	24hr (less idle time)	24hr (more idle time)
Energy consumption (kWh/day)	21.3	17.1
Dome valve activations	27	16
Estimated-ash throughout (kg)	110	108

Fig: 7.2 OPTIMIZATION COMPARISION

Source: SCADA export data and logbook entries from Ponni Sugars (Erode) Cogeneration Unit.

7.3 ECONOMIC IMPACT AND ENERGY SAVINGS

Based on the trial results, extrapolated energy savings for all six transporters operating on ESP ash hoppers were calculated.

Summary of Savings Potential:

- i. Energy savings = ~ 4.2 kWh/unit/day
- ii. Number of transporters = 6
- iii. Total daily savings = ~ 25.2 kWh/day
- iv. Annual savings = $\sim 9,198$ kWh/year

Monetary Savings:

- i. Considering industrial electricity rate at ₹7.5/kWh (Tamil Nadu EB rates)
- ii. Estimated savings = ₹68,985/year

This projection does not include savings from reduced wear on compressor motors, dome valves, and reduced maintenance cost due to lower mechanical cycles.

7.4 ENVIRONMENTAL AND OPERATIONAL BENEFITS

Environmental improvements from reduced compressor usage and controlled ash discharge cycles include:

- i. Lower noise levels from air bursts and motors.
- ii. Reduced CO₂ emissions from power generation (approx. 0.9 kg CO₂ per unit saved).
- iii. Stabilized ash flow reduces potential for particulate leakage.
- iv. Enhanced safety due to controlled valve actuation and reduced pressure shock

REFERENCE:

- Central Pollution Control Board Guidelines on Emissions
- https://cpcb.nic.in/uploads/National_Ambient_Air_Quality_Standards.pdf

7.5 DISCUSSION ON SCALABILITY AND INDUSTRY IMPLICATIONS

The trial demonstrated that a simple retrofit involving pressure transmitters and logic control modification can lead to meaningful gains in energy conservation and system responsiveness. Considering the uniformity of ash handling logic across industries, this model can be applied to:

Cement kiln ESP systems
Biomass-fired industrial boilers
Coal-based thermal power plants

Scalability depends on the control system (PLC/SCADA) flexibility and sensor durability. The cost-to-benefit ratio is positive even for small-scale operations.

Sources:

“Smart Controls in Ash Handling”, Journal of Clean Energy Systems, 2020
Yokogawa and Siemens Ash Control Case Studies

7.6 CHALLENGES ENCOUNTERED AND RESOLUTIONS

Several challenges were faced during the trial:

Challenge 1 – Pressure Signal Instability:

Cause: Moisture in compressed air line.

Solution: Installation of heater and use of shielded cable for analog signal.

Challenge 2 – Calibration Drift:

Cause: High temperature during mid-day shifts.

Solution: Implement auto-zero correction logic in PLC code.

Challenge 3 – Operator Familiarization:

Cause: Manual override habit persisted.

Solution: Training and visual indicator panel showing live pressure values.

Each of these issues was addressed and documented, ensuring that the system remained stable and reproducible.

7.7 FINAL REMARKS AND FUTURE SCOPE

The project validates the role of real-time instrumentation in improving auxiliary system behavior in industrial cogeneration plants. With an emphasis on minimal intervention, high-impact retrofits like pressure-based transporter control offer a significant advantage.

Future Scope Includes:

- i. AI-based compressor demand forecasting based on transporter fill level patterns.
- ii. Integration with cloud-based reporting dashboards.
- iii. Expansion to bagasse handling, soot blower scheduling, and water pumping.

This project forms the basis for a digital twin model of auxiliary systems, supporting India's mission for smart manufacturing.

7.8 REFERENCES

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